

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Geology

Questions: 15

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

1. Answer the following within 150 words each : 8x5=40 What is a meteorite? Describe different types of meteorites. Discuss the spectral and spatial resolution of a satellite image. What is a moraine? Discuss different types of moraines. Discuss five criteria to recognise the top of a bed by primary features. Describe strike, dip direction and dip amount of a bed. Can the strike and dip be measured in all types of orientation of beds? How are strike, dip direction and dip amount of horizontal, tilted and vertical strata represented on a map?

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2. [UNKNOWN] paper-i

2. (a) What is a volcano? What are the different types of volcano based on the types of eruption and places where the eruption occurred? 15 Explain in detail about panchromatic, multispectral and hyperspectral remote sensing images with example. 10 What is a fold? Draw neat sketches and discuss Ramsay's classification of fold based on dip isogon. 15

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3. [UNKNOWN] paper-i

3. (a) What are the differences between weathering and erosion? Discuss four different types of mechanical weathering. 10 What is a fault? Draw a neat diagram and discuss different terminologies associated with a fault plane. Discuss four criteria for recognising fault in a field. 15 What is an earthquake? Discuss the causes and effects of an earthquake. List the major geographic distribution of earthquakes worldwide. 15

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4. [UNKNOWN] paper-i

4. (a) What is isostasy? Discuss three different theories of isostasy. What are the merits and limitations of these theories? 10 Discuss four erosional and four depositional landforms associated with a river. 15 Discuss about stereographic projections for structural analysis with neat diagrams. 15 FSI—P—GLY /46 2 ===== PAGE 3 ===== SECTION-B

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5. [UNKNOWN] paper-i

5. (a) What is orthogenesis? Comment on the progressive trend of evolution from Hyracotherium to Equus. 8 How are abiotic conditions reflected in the test ornamentation of microfossils? Enumerate with example. 8 What is palaeogeography? How does the present petrographic domain reconstruct the past events? 8 With the help of a neat sketch, draw a labelled diagram of hydrological cycle. 8 What are creep, solifluction and landslide? Enumerate their significance that leads to mass scale devastation. 8

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6. [UNKNOWN] paper-i

6. (a) Discuss about the two-fold classification of Gondwana group of rocks, mentioning the stages and their ages. Comment on the important floras of Lower Gondwana and Upper Gondwana basins. 15 What do you mean by stratigraphic boundary problems? How do you substantiate the observation on Pakhal Controversy? 10 Define the terms Rock Quality Designation and Rock Mass Rating (RQD 85 RMR). How are these parameters useful in the analysis of the design of roof support system in tunnel excavations? 15

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7. [UNKNOWN] paper-i

7. (a) What is 'dimorphism' in foraminiferal reproduction? How is dimorphism established through the features in foraminiferal tests? Discuss the evolutionary changes of foraminifers through geological ages. 15 In Indian stratigraphy, Cuddapah system plays a dominant role. Attempt the stratigraphic classification of the Cuddapah group of sediments. Add a note on the age of these rocks with special reference to the economic potential associated. 15 What do you understand by the terms Hydraulic Conductivity and Storage Coefficient? 10 FSI—P—GLY /46 3 P.T.O. ===== PAGE 4 =====

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8. [UNKNOWN] paper-i

8. (a) Which palaeogeographic condition is generally indicated by the presence of brachiopods? How is 'habitat shifting' represented by brachiopods in rocks throughout time? Discuss about major features of brachiopod shells. 15 What is Himalayan Orogeny? Briefly describe the structure and tectonic history of Himalayas. 15 What is rainwater harvesting? Enumerate the different types of structures constructed for rainwater harvesting. 10 * * * FSI-P-GLY /46 4 BS9-594

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9. [UNKNOWN] paper-ii

Q2. (a) What are primary sedimentary structures ? Write a note on the significance of ripple marks in terms of sedimentary environment. 15 Explain the concept of metamorphic facies with suitable diagram and write a note on metamorphic facies related to contact metamorphism. 15 What is eutectic crystallization ? Show the behaviour of eutectic crystallization in the binary system, giving appropriate example. 10

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10. [UNKNOWN] paper-ii

Q3. (a) What are biaxial interference figures ? Explain how interference figures are obtained under conoscopic condition and add a note on types of biaxial interference figures. 15 Write an elaborate note with neat diagram on mineralogical composition and texture of common mafic and felsic igneous rocks. 15 Elaborate the classification scheme and petrographic characteristics of sandstone. 10

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11. [UNKNOWN] paper-ii

Q4. (a) Explain the symmetry configuration of 'Normal Class' of 'Orthorhombic System' and discuss the characteristic forms. 15 How are amphibole and pyroxene distinguished based on crystal structures and optical properties? 10 Write mineral composition and texture of the following rocks — charnockite, phosphorite and limestone. 15 FSI-D-GLY 2 ===== PAGE 3 ===== SECTION B

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12. [UNKNOWN] paper-ii

Q5. (a) How are early and late magmatic ore deposits distinguished ? Give your answer with suitable examples. 8 What are the salient features of 'National Mineral Policy' ? Discuss with special emphasis on 'Strategic Minerals'. 8 Write briefly about classification and geochemical distribution of the elements in the Earth's crust. 8 Write about the source, speciation and fixation of 'Arsenic Contamination' in groundwater, with special emphasis to the Indian context. 8 Assuming the specific gravity of 4.1 and average grade of 54.1% Fe, calculate the total tonnage of iron ore and metallic iron content in a 10 m thick and 50 m wide iron ore body with a strike length of 500 m. 8

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13. [UNKNOWN] paper-ii

Q6. (a) What is coalification process and how is it related to different ranks of coal? 15 What are the different methods of sampling practised during mineral exploration and mining ? Give reasons for choice of sampling method followed in different types of ore deposits. 15 What is acid mine drainage ? Briefly explain its environmental consequences and controlling measures. /0

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14. [UNKNOWN] paper-ii

Q7. (a) Elaborate the process of origin, migration and trapping conditions for hydrocarbon. 15 Write the salient points of interpretation of electrical resistivity data and discuss suitability of the electrical resistivity method for groundwater exploration. 15 Give a brief account of mineral resources in 'Exclusive Economic Zone' (EEZ) of India. 10 FSI-D-GLY 3
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15. [UNKNOWN] paper-ii

Q8. (a) Write a note on geographical distribution, mineralogy, mode of occurrence and origin of lead-zinc (Pb — Zn) ore deposits in India. 15 Write a note on methods of radioactive waste disposal with special reference to geological parameters. 15 Distinguish between 'Chondrite' and 'Achondrite' and describe the 'Carbonaceous Chondrite'. 10 FSI-D-GLY 4

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